

One-Page Individual Reflection

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During this assignment, I worked on the design and development of a Hotel Accommodation and Reservation Management System using C++, modelling guests, rooms, reservations, and services as separate classes linked through inheritance. I went beyond a simple room hierarchy by combining a Deluxe room and a Spa service into one bookable PremiumPackage class through multiple inheritance, which meant both branches had to share a common HotelItem base. Making Room and Service inherit HotelItem virtually, instead of normally, was the key decision that avoided the diamond-inheritance duplication problem the assignment asked us to address.

A major challenge was a subtle lexical bug: a comment in my source code accidentally contained the character sequence that closes a C++ block comment early, so the compiler tried to parse plain English as code and produced a wall of unrelated errors. Tracing this back taught me to trust the first error in a long list rather than the last. A second challenge came while writing the PremiumPackage constructor: my first attempt tried to initialise Room directly, even though only DeluxeRoom and SpaService are its direct bases. Fixing this clarified how virtual base initialisation shifts to the most-derived class in C++.

The assignment improved my understanding of encapsulation, abstraction, inheritance (single, hierarchical, multilevel, multiple, and hybrid with a virtual base class), pointers, virtual functions, runtime polymorphism, and STL containers. Testing situations such as valid reservations, an already-occupied room, invalid reservation IDs, service selection, cancellation, and non-numeric input helped me see why careful validation matters in real software.

The project connects with SDG 8 and SDG 9, since a reliable, extensible reservation system reduces manual front-desk workload and billing errors for small hospitality businesses, and shows how inheritance-based software can scale with changing service offerings. Overall, the assignment helped me connect C++ classroom concepts with a practical application and improved my problem-solving, coding, testing, and documentation skills, contributing to the mapped course outcomes on object-oriented design and validation.